

# PRIYANKA RAJAN

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## EDUCATION

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### University at Buffalo & Roswell Park Comprehensive Cancer Center, Buffalo, NY, USA

Ph.D. in Cancer Sciences

2019 - 2025

Thesis focus: Immunotherapy and targeted therapy in breast cancer

M.S. in Oncology and Cancer Biology

2017 - 2019

Thesis focus: Cellular response to chromatin targeted anti-cancer agents

### PSG College of Technology, Coimbatore, India

2013 - 2017

B. Tech. in Biotechnology

Thesis focus: Molecular mechanisms of cell division

## EXPERIENCE

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### Independent Researcher

Apr 2025 - present

- [Tracing cellular and tissue origins of plasma protein biomarkers for early cancer detection using ML on single-cell RNA-seq and spatial omics data.](#)
- [Building a tumor immune-suppression signature predictive of immunotherapy response and pre-metastatic priming.](#)
- [Quantifying BRCA variant frequencies across Indian sub-populations using the GenomeIndia dataset.](#)
- [Developing Medindicator, an app that flags drug-drug interactions using established interaction data.](#)

### Roswell Park Graduate Division, Buffalo, NY

Aug 2019 - Feb 2025

Graduate Research Assistant

- Led the design, implementation, and execution of all pre-clinical laboratory mouse models & tissue-cell culture experiments.
- Assisted in the design, implementation of clinical trial (NCT04511039), patient PMBC processing and PDX study development
- Managed daily laboratory activities including the maintenance of data, inventories & placing orders; planned, performed and analyzed experiments; reviewed scientific literature; proposed & tested treatment strategies with medical & research professionals; mentored younger trainees.
- Wrote, reviewed and edited grant applications, scientific publications, and presented at international conferences.
- Designed and delivered lectures to incoming grad students, summer undergraduate and high school students on Gene manipulation techniques, Tumor Microenvironment and Metastasis, How to present scientific data. Held office hours and prepared & graded exams.

### Enhanced Pharmacodynamics LLC, Buffalo, NY

Jan 2019 - May 2019

Senior Research Aide

- Assisted in development of a web application to assess immune toxicity of an anti-cancer drug candidate from phase I clinical trial data based on existing population pharmacokinetic/pharmacodynamic (PK/PD) models.

## TECHNICAL SKILLS

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**Computational & Data Analysis:** R, Python, Linux/Bash, RNA-seq and scRNA-seq data analysis

**Protein & Cell Biology:** Multiparametric flow cytometry (12 color panels), primary cell culture, western blot, proliferation, migration, comet, cell death assays, reporter assays

**Molecular Biology & Engineering:** CRISPR-Cas9, RT-qPCR, cloning, primer design, DNA/RNA isolation, bulk RNA-seq, scRNA-seq

**In Vivo Models:** Syngeneic and patient-derived xenograft (PDX) mouse models, preclinical drug efficacy testing

## RESEARCH PROJECTS

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### **Blockade of p38 reverses the immune suppressive tumor microenvironment in breast cancer** *Jun 2020 – Feb 2025*

- Investigated how tumor suppresses the immune response focusing on the role of tumor p38 MAP kinase using p38 inhibitor (Ralimetinib) and p38 knockout (CRISPR-Cas9) strategies in pre-clinical breast tumor models. Demonstrated that tumor p38 mediates MDSC and TAM recruitment via cytokines & EVs and p38 blockade reprograms immune-suppressive myeloid cells. Further demonstrated that p38 blockade reverses T cell exhaustion, enhancing activation and producing synergistic effects on tumor growth and metastasis in combination with anti-PD-1 therapy. (5 abstracts, 1 manuscript under review).
- Supervisor: Dr. Andrei Bakin, Roswell Park Comprehensive Cancer Center, Buffalo NY.

### **Development of a two-drug combination therapy for p53 mutant cancers**

*Aug 2020 – Feb 2025*

- In collaborative effort, I developed a two drug strategy consisting of TAS102 and PARP inhibitor for selectively targeting p53 mutant cancers (NCT04511039). Determined mechanism of action and efficacy of the combination therapy in in vitro models and pre-clinical CDX & PDX models. Developed a triple combination therapy with immunotherapy (6 abstracts, 1 manuscript published and 1 manuscript under review).
- Supervisor: Dr. Andrei Bakin, Roswell Park Comprehensive Cancer Center, Buffalo NY.

### **Comparison of p53 response to DNA and chromatin damage**

*Jan 2018 – Aug 2019*

- Determined cellular response to Curaxin, a novel anti-cancer small molecule that destabilizes DNA histone assembly, focusing on role of p53 using p53 wildtype & mutant isogenic cell lines. Developed comet assay using raw lab ingredients that brought down cost 30x and improved robustness. Demonstrated that curaxin activates p53 targets not via DNA damage but via a broader opening up of chromatin (2 abstracts and 1 article published in Nucleic Acids Research).
- Supervisor: Dr. Katerina Gurova, Roswell Park Comprehensive Cancer Center, Buffalo NY.

### **Investigation of role of *but2* in fission yeast cytokinesis**

*Dec 2016 – May 2017*

- Assessed the role of *but2* in regulating fission yeast separation to two daughter cells. Generated reporter strains by crossing *but2* deletion with strains carrying cell division genes tagged to fluorescent reporters. Measured effects using time-lapse epifluorescence microscopy.
- Supervisor: Dr. Mithilesh Mishra, Tata Institute of Fundamental Research (TIFR), India

### **Assessment of role of nutraceuticals in cancer chemosensitization**

*May 2016 – Jul 2016*

- Assessed the role of >40 nutraceuticals in cancer chemosensitization by literature survey. Co-authored a book chapter on the role of Berberine in sensitizing cancer to chemotherapy in the book series - Role of Nutraceuticals in Cancer Chemosensitization.
- Supervisor: Dr. Alok Chandra Bharti, University of Delhi, India.

### **Analysis of promoter proximal pausing in the human genome**

*Jul 2015 – Nov 2016*

- Led a team of 3 to determine if promoter proximal pausing is influenced by motifs. Analyzed GRO-seq data to identify and characterize transcriptionally paused genes. Evaluated GC richness, presence of G4, R-loops forming sequence and tandem repeats (2 abstracts presented).
- Supervisor: Dr. V. Ramamurthy, PSG College of Technology, Coimbatore, India.

## ACHIEVEMENTS

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- Completed Claude Science Hackathon (1 of 150 researchers worldwide) 2026
- Awarded for student academic excellence 2025
- Awarded GSA Conference Fund 2022
- Awarded UB Mark Diamond Research Fund 2021
- Selected as Career Experience fellow by SUNY Research Foundation 2019
- Qualified Indian national *Graduate Aptitude Test in Engineering* (GATE) with 98.2 percentile (Rank 134) 2017
- Selected for a 6-month thesis work at the Tata Institute of Fundamental Research (TIFR), India 2016
- Selected among the top 1% applicant in India for Summer Research Fellowship program by Indian Academy of Science (IAS) 2016
- Best Paper Award, Ramanujan Association for Mathematics, PSG Tech. 2015, 2016.

## PUBLICATIONS

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### Research articles

- Rajan, P.**, Zollo, R., Guo, Y., Lieberman, M., Alruwaili, M.M., Zonneville, J., Morreale, B., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. Targeting p38 reprograms the tumor immune microenvironment to inhibit metastatic breast cancer. *bioRxiv* 2025.05.08.652949. Under review at NPJ Precision Onc.
- Alruwaili, M.M., Guo, Y., Zonneville, J., Melendy, T., Straubinger, R., Foster, B.A., **Rajan, P.**, Withers, H., Chatley, S., Iyer, R., Fountzilias, C., Bakin, A.V. A Sequential Triple-Drug Strategy for Selective Targeting of p53-Mutant Cancers. *bioRxiv* 2025.11.06.687028. Under review at *Can Lett*.
- Luzhin, A., **Rajan, P.**, Safina, A., Leonova, K., Stablewski, A., Wang, J., Robinson, D., Isaeva, N., Kanditze, O., Gurova, K. (2023). Comparison of cell response to chromatin and DNA damage. *Nucleic Acids Research*. doi:10.1093/nar/gkad865.
- Alruwaili, M.M., Zonneville, J., Naranjo, N., Serio, H., Melendy, T., Straubinger, R., Gillard, B., Foster, B.A., **Rajan, P.**, Attwood, K., Chatley, S., Iyer, R., Fountzilias, C., Bakin, A.V. A Synergistic Two-Drug Therapy Specifically Targets a DNA Repair Dysregulation That Occurs in p53-Deficient Colorectal and Pancreatic Cancers. (2023). *Cell Reports Medicine*. doi:10.1016/j.xcrm.2024.101434.
- Guo Y., **Rajan P.**, Alruwaili M., Barbi J., Abrams S., Bakin A. Novel immunotherapy combination for p53 mutant metastatic breast cancer leveraging weaknesses in the DNA damage response. Under preparation.

### Review articles & Book chapters

- Rajan P.**, Bakin A. p38 as a mediator of tumor-immune crosstalk in breast cancer. Under preparation.
- Bharti A.C., **Rajan P.**, Jadli M., Pande D., Singh T., Bhat A. (2018). Berberine as an Adjuvant and Sensitizer to Current Chemotherapy. In Bharti A.C., Aggarwal B. B. (Eds.), *Cancer Sensitizing Agents for Chemotherapy- Role of nutraceuticals in cancer chemosensitization* (Vol. 2, 221-240, 2017), Elsevier.

## ORAL PRESENTATIONS

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- American Association for Cancer Research, Apr 17-22, 2026, San Diego, CA.
  - Alruwaili, M.M., Guo, Y., Zonneville, J., Melendy, T., Straubinger, R., Foster, B.A., **Rajan, P.**, Withers, H., Chatley, S., Iyer, R., Fountzilias, C., Bakin, A.V. A novel therapeutic approach for targeting p53-mutant cancers by leveraging DNA damage response vulnerabilities. Apr 21, 2026.
- Translational Research Cancer Centers Consortium, Feb 19-21, 2025, Seven Springs, PA.
  - Rajan, P.**, Zollo, R., Guo, Y., Lieberman, M., Alruwaili, M., Zonneville, J., Morreale, B., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. Targeting p38 to reverse an immune suppressive tumor microenvironment in metastatic breast cancer. Feb 20, 2025.
  - Guo Y., **Rajan, P.**, Alruwaili M., Alqarni M., Barbi J., Abrams S., Bakin A. Novel immunotherapy combination for p53 mutant metastatic breast cancer leveraging weaknesses in the DNA damage response. Feb 19, 2025.

## POSTERS

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1. AACR (General Meeting), Apr 25-30, 2025, Chicago, IL.
  - Bakin, A., Guo, Y., **Rajan, P.**, Alruwaili, M.M., Barbi, J., Abrams, S., Fountzilas, C. Abstract 663: A novel combination immunotherapy for p53 mutant metastatic breast cancer leveraging vulnerabilities in the DNA damage response. *Cancer Res* (2025) 85 (8\_Supplement\_1): 663.
  - Guo Y., **Rajan P.**, Alruwaili, M.M., Alqarni, M.A., Barbi, J., Abrams, S., Bakin, A. Abstract 5991: Novel triple-drug combination strategy for p53 mutant cancers leveraging their DNA damage response liabilities. *Cancer Res* (2025) 85 (8\_Supplement\_1): 5991.
2. Translational Research Cancer Centers Consortium, Feb 19-21, 2025, Seven Springs, PA.
  - **Rajan, P.**, Zollo, R., Guo, Y., Lieberman, M., Alruwaili, M.M., Zonneville, J., Morreale, B., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. Targeting p38 to reverse an immune suppressive tumor microenvironment in metastatic breast cancer. Feb 20, 2025.
  - Guo Y., **Rajan P.**, Alruwaili, M.M., Alqarni, M.A., Barbi, J., Abrams, S., Bakin, A. Novel immunotherapy combination for p53 mutant metastatic breast cancer leveraging weaknesses in the DNA damage response. Feb 19, 2025.
3. San Antonio Breast Cancer Symposium, Dec 10-13, 2024, San Antonio, TX.
  - Guo Y., **Rajan P.**, Alruwaili, M.M., Alqarni, M.A., Barbi, J., Abrams, S., Bakin, A. Abstract P3-08-27: Novel immunotherapy combination for p53 mutant metastatic breast cancer leveraging weaknesses in the DNA damage response. *Clin Cancer Res* (2025) 31 (12\_Supplement): P3-08-27.
4. AACR (General Meeting), April 5-10, 2024, San Diego, California.
  - **Rajan P.**, Zollo, R., Lieberman, M., Guo, Y., Alruwaili, M.M., Alqarni, M.A., Morreale, B., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. (2024). Abstract 5536: The role of p38 MAPK in the tumor-induced immune suppressive microenvironment in metastatic breast cancer. *Cancer Res*, 84(6 Supplement), 5536-5536.
  - Alruwaili, M.M., Naranjo, N., **Rajan P.**, Jo, K., Kim, D.-K., Melendy, T., Straubinger, R., Fountzilas, C., Bakin, A. (2024). Abstract 7590: Novel therapeutic approach for targeting p53 mutant colorectal cancers by affecting post-replicative DNA repair. *Cancer Res*. 84(6 Supplement), 7590-7590.
5. Roswell Park Scientific Retreat, Nov 2-3, 2023, Niagara Falls, NY.
  - **Rajan P.**, Zollo, R., Lieberman, M., James, C., Sharda, A., Guo, Y., Morreale B., Alruwaili M., Alqarni M., Olejniczak S., Barbi J., Abrams, S., Bakin, A. (2023). Abstract 19: Inhibition of p38 MAPK reduces the tumor-induced immune suppressive microenvironment in metastatic breast cancer.
6. AACR (General Meeting), April 14-19, 2023, Orlando, Florida.
  - **Rajan, P.**, Zonneville, J., Zollo, R., Honikel, M., Raudins, S., Colligan, S., Morreale, B., Alruwaili, M., Alqarni, M., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. (2022). Abstract 73: Blockade of p38 MAPK reduces the tumor-induced immune suppressive microenvironment in metastatic breast cancer. *Cancer Res*, 83(7 Supplement), 73-73
  - Alruwaili, M. M., Zonneville, J., Alqarni, M. A., **Rajan, P.**, Serio, H., Straubinger, R., Fountzilas, C., Bakin, A. (2023). Abstract 3397: Evaluation of a novel two-drug combination strategy for p53-deficient colorectal and pancreatic cancers. *Cancer Res*. 83(7 Supplement), 3397-3397.
7. AACR Special Conference: Pancreatic Cancer, September 13-16, 2022, Boston, MA
  - Alruwaili, M. M., Zonneville, J., Alqarni, M. A., **Rajan, P.**, Serio, H., Straubinger, R., Fountzilas, C., Bakin, A. (2022). Abstract B039: Development of a selective therapeutic intervention for p53 mutant for pancreatic adenocarcinoma. *Cancer Res*. 82(22 Supplement), B039-B039.

8. Gordon Research Conference on Chromatin Structure and Function, May 29-Jun 3, 2022, Barcelona, Spain.
  - Safina, A., Luzhin, A., **Rajan, P.**, Leonova, K., Kantidze, O., Gurova, K. (2022, May). Response to chromatin damage in normal and tumor cells.
9. San Antonio Breast Cancer Symposium, Dec 7-10, 2021, San Antonio, TX.
  - **Rajan, P.**, Zonneville, J., Colligan, S., Abrams, S., Bakin, A. (2022). Abstract P5-17-06: P38 kinase as a therapeutic target to reverse an immune suppressive tumor microenvironment in metastatic breast cancer. Cancer Res, 82 (4 Supplement): P5-17-06.
10. 5th Annual Genome day, New York State Center of Excellence in Bioinformatics & Life Sciences; January 2019; Buffalo, NY.
  - **Rajan, P.**, Safina, A., Gurova, K.V. Damage to nucleosomes is sensed by p53.

## GRANTS AND FELLOWSHIPS

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1. University at Buffalo Mark Diamond Research Fund *Sep 2021 - Oct 2022*
  - Grant Number: FA2106
  - Title: Combining p38 blockade with Immune Checkpoint Inhibition in Triple Negative Breast Cancer
  - Role on project: Principal Investigator
2. SUNY Research Foundation Career Experience Fellow *Jan 2019 - May 2019*
  - Selected for a University at Buffalo sponsored program that gives students the opportunity to work for life sciences and advanced manufacturing companies in the Western New York area.
  - Supported my work at Enhanced Pharmacodynamics, Buffalo.
3. Indian Association of Sciences Summer Research fellowship *May 2016 - July 2016*
  - A national science academy funded two month summer fellowship to enable students to work with scientists associated with the academy.
  - Supported my work at Dr. Bharti's lab at University of Delhi.
4. All India Vijaya Bank Officers Association scholarship *Jun 2013 - May 2017*
  - Scholarship awarded to outstanding high school students in support of college tuition
5. Vijayalakshmi Trust scholarship *Jun 2013 - Dec 2013*
  - Scholarship awarded to outstanding high school students in support of first semester college expenses

## PROFESSIONAL AFFILIATIONS

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- International Society for Computational Biology *Oct 2025 – present*
- American Association for Cancer Research *Jan 2022 – present*
- Society for Cancer Immunotherapy ConnectED *Apr 2022 – present*
- UB graduate student association *Aug 2017 – Feb 2025*
- Roswell park graduate student association *Aug 2017 – Feb 2025*
- Biotechnology association, PSG College *Jun 2013 – May 2017*

## SERVICE

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### Activities

- Write blog posts on topics in computational biology and biomedical research on [Substack](#) (2026-present)

- Ad hoc peer reviewer for multiple journals including *Comms Bio* and *Can Immunol Immunotherapy* (2026–present)
- Co-organized the Biomedical Career Fair at Roswell Park (2023); recruited speakers from Complete Genomics and STEMCELL Technologies and managed all social media outreach for the event
- Designed and launched the [website](#) for the *Bakin lab* (2022), serving as the lab's primary public-facing digital presence
- Authored a summary report on invited speaker talks for SCI-ROI's annual event as a volunteer writer for the nonprofit (2022)
- Judged undergraduate research presentations in the Roswell Park Summer Program (2021–2022)
- Introduced cancer research themes to 40 high school and undergraduate students on the Buffalo Healthcare Explorations (BHE) Career Day, using an interactive Jeopardy!-style format (2021–2022)
- Reviewed graduate student grant applications as a council member for the Mark Diamond Research Fund, University at Buffalo (2021)
- Raised funds for cancer research through the Ride for Roswell (2019–2024)

## Teaching

- Designed and delivered lecture on Gene Manipulation techniques as part of Techniques Associated with Cancer Sciences (TACS) course to the incoming graduate students
- Served as Teaching Assistant for Integrated Cancer Sciences: Tumor Microenvironment and Metastasis course. Held review classes, prepared exam material and assisted with grading

## Mentorship

### *Graduate trainees*

- Ashley Guo, MS Bioengineering, Cornell University. Research Technician *Sep 2023 – Feb 2025*
- Erica Norris, MS Biotechnology, Duquesne university. Summer Trainee *Jun 2024 – Aug 2024*
- Natsumi Naranjo, MS Biological Sciences, University at Buffalo. Research Technician *Sep 2023 - Apr 2024*
- Georgia Kyriacou, MD, University at Buffalo. Summer Trainee *May 2023 - Aug 2023*
- Mohammed Alqarni, MS Medical Laboratory sciences, University of Rhode Island. Rotation student *Aug 2022 – Oct 2023*

### *Undergraduate Trainees*

- Sofija Raudins, BS Biochemistry, Kent State University. Summer Trainee *May 2022 – Aug 2022*
- Megan Melnick, BS Biological Sciences, University at Buffalo. Undergraduate student *Jun 2020 – Feb 2020*
- Dante Poe, BS Biochemistry and Molecular biology, Michigan State University. Summer Trainee *Jun 2018 – Aug 2018*